
Topological groups

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It turns out that, in many cases, the study of a topological group in a neighbourhood of the neutral element will give valuable information on the whole group

Theory of Lie Groups
Chevalley

I am writing these notes for my presentation at the Harmonic Analysis and Operator Algebras seminar at CIMAT-Merida.

1 Quotient groups

Let G be a topological group and H a subgroup of G . Let G/H denote the set of all cosets $xH = \{xh : h \in H\}$ and $\varphi : G \rightarrow G/H$ the canonical mapping. We might induce a topology on G/H say that a set $A \subset G/H$ is open if and only if $\varphi^{-1}(A)$ is open in G . Note that $A = \{aH : a \in A\}$ and therefore a set A is open if and only if $\bigcup_{a \in A} aH$ is open in G .

Theorem 1.1. *Let G be a topological group and H be a subgroup of G . Let G/H be the quotient space, endowed with the quotient topology, and φ the canonical map from G into G/H . Then:*

1. φ is onto;
2. φ is continuous;
3. φ is open.

Theorem 1.2. *Let G be a topological group and H be a subgroup of G . Then G/H is a T_1 space if and only if, H is closed.*

Proposition 1.3. *Let G be a topological group and H be a subgroup of G . Then G/H is a discrete topological space if, and only if, H is open.*

So far, we have not proven that this quotient is a group, the next proposition tell us that the quotient G/H is a topological group if H is a normal subgroup (invariant under conjugation).

Theorem 1.4. *Let G be a topological group and H be a normal subgroup of G . Then*

1. *The canonical mapping $\varphi : G \rightarrow G/H$ is a continuous and open homomorphism.*
2. *G/H , endowed with the quotient topology is a topological group.*
3. *If G is a topological group, the G/H is completely regular topological group if, and only if, H is a closed subgroup of G .*

Remember that a topological space is said to be completely regular if for any closed set A and any point $x \in X \setminus A$, there exist a real continuous $f : X \rightarrow \mathbb{R}$ function such that

$$f(x) = 1 \quad \text{and} \quad f|_A = 0$$

To better understand the properties of the quotient group, we turn to another key result.

Theorem 1.5. *Let G be a topological group and H be a normal closed subgroup of G . Then:*

1. G/H is compact if G is;
2. G/H is locally compact if G is locally compact topological group;
3. G/H satisfies the first axiom of countability if G does;
4. G/H satisfies the second axiom of countability if G does;

Finally we consider the next results about compact subgroups

Proposition 1.6. *Let H be a compact normal subgroup of a topological group G . Then the canonical mapping $\varphi : G \rightarrow G/H$ is closed.*

Corollary 1.7. *Let H be a normal subgroup and M any closed subgroup of a topological group such that $H \subset M$. Then $\varphi(M)$ is a closed subgroup of G/H .*

1.1 a bit of homogeneous spaces

While the requirement of normality is quite restrictive, requiring H to be merely a closed subgroup still allows for the construction of significant geometric structures, such as homogeneous spaces.

Theorem 1.8 (Proposition 2.2[Fol94]). *Suppose H is a subgroup of a topological subgroup G*

1. *If H is closed then G/H is Hausdorff.*
2. *If G is locally compact, so is G/H .*

Definición 1.9 (G -spaces). *Let G be a locally compact Hausdorff group and S be a locally compact Hausdorff space. The space S is called a G -space if there exist a continuous function $\varphi : G \times X \rightarrow X$, which we will denote by $(g, x) \mapsto g \cdot x$, with the next properties*

- $g_1 \cdot (g_2 \cdot x) = (g_1 g_2) \cdot x \quad (\forall x \in X) \quad \text{and} \quad (\forall g_1, g_2 \in G),$
- $1 \cdot x = x \quad (\forall x \in X).$

and

$$\begin{aligned} \varphi_g : X &\rightarrow X \\ x &\mapsto g \cdot x \end{aligned}$$

be a homeomorphism for any $g \in G$.

Ejemplo 1. The standard example of G -space is the quotient G/H endowed with the quotient topology induced by the canonical map, where H is a closed subgroup and G acts by left multiplication.

Almost any example is one of this In fact let S be a G -space, pick some $s_0 \in S$ and define a $\phi : G \rightarrow S$ by $\phi(x) = x \cdot s_0$ and let $H = G_{s_0}$. Then H is a closed subgroup of G and ϕ is a surjective continuous of G onto S that is constant in the left cosets, then there exist a continuous bijection $\Phi : G/H \rightarrow S$ such that the next diagram commutes

$$\begin{array}{ccc} G & \xrightarrow{\phi} & S \\ \downarrow \pi & & \uparrow \Phi \\ G/H & & \end{array}$$

The only additional condition for the identification of S with G/H is the continuity of the map Φ^{-1} . This is not always the case.

Theorem 1.10. *With the notation as above, if G is a σ -compact then Φ is a homeomorphism.*

Remember that a topological space is said to be σ -compact if it is the union of countably many compact subspaces.

In the special case when G is a Lie group and H is a closed subgroup of G (and therefore a Lie group) we can give a smooth structure to the quotient G/H

Proposition 1.11 (Proposition 4.1 [Arv03]). *Let G be a Lie group and H be a closed subgroup of G . Then there is a unique way to make G/H a manifold such that the projection $\pi : G \rightarrow G/H$ is a submersion; that is, $d\pi_g$ is onto for all $g \in G$*

It is important to remember that in this case the group H is not necessarily a normal subgroup

Theorem 1.12 (Proposition 4.2 [Arv03]). *Let $G \times M \rightarrow M$ be a transitive action of a Lie group G on a manifold M , and let $K = G_m$ be the isotropy group of a point m . Then*

1. *The subgroup K is a closed subgroup of G .*
2. *The natural map*

$$\begin{aligned}\pi : G/K &\rightarrow M \\ gK &\mapsto g \cdot m\end{aligned}$$

is a diffeomorphism.

3. *The dimension of G/K is $\dim G - \dim K$.*

Ejemplo 2 (Spheres). There are several Lie groups that act transitively over the sphere S^n , just as an example we can consider $\mathrm{SO}(n+1), \mathrm{O}(n+1)$

$$S^n = \mathrm{SO}(n+1)/\mathrm{SO}(n) \cong \mathrm{O}(n+1)/\mathrm{O}(n)$$

$$S^{2n+1} \cong \mathrm{SU}(n+1)/\mathrm{SU}(n) \cong \mathrm{U}(n+1)/\mathrm{U}(n)$$

Ejemplo 3 (Projective spaces). The real projective space and the complex projective space are diffeomorphic to quotients of Lie groups

$$\mathbb{R}P^n \cong \mathrm{SO}(n+1)/\mathrm{O}(n) \cong \mathrm{O}(n+1)/\mathrm{O}(n) \times \mathrm{O}(1)$$

$$\mathbb{C}P^n \cong \mathrm{SU}(n+1)/\mathrm{U}(n) \cong \mathrm{U}(n+1)/\mathrm{U}(n) \times \mathrm{U}(1)$$

Returning to the general results in *Introduction to topological groups*[Hus18]

2 Products of topological groups

Theorem 2.1. *Let A be an index set. For each $\alpha \in A$, let G_α be a topological group. Then*

$$G = \prod_{\alpha \in A} G_\alpha$$

endowed with the product topology, is a topological group.

Theorem 2.2. *Let $G = \prod_{\alpha \in A} G_\alpha$ be the direct product of topological groups endowed with the product topology. Then:*

1. *G is compact if, and only if, each G_α is*
2. *G is Hausdorff topological group if, and only if, each G_α is*
3. *G is locally compact if all G_α are compact topological group except for a finite number of α_i 's, and for this α_i 's the group G_{α_i} is locally compact.*

3 Locally compact topological groups

A brief collection of important results in locally compact topological groups. This is a more general case of a Lie group.

Theorem 3.1. *Every topological group is locally compact if, and only if, there exist a compact neighborhood of the identity*

Theorem 3.2. *Every locally compact topological group is normal*

Corollary 3.3. *Every locally compact subgroup of a topological group is a closed subgroup*

Theorem 3.4. *Every compact open neighborhood U of the identity of a topological group contains a compact, open, and closed subgroup*

Theorem 3.5. *The component C of the identity e of a locally compact topological group G is the intersection of all open subgroups of G*

and our final theorem

Theorem 3.6. *Let G be a locally compact topological group. The following statements are equivalent:*

1. G is connected;
2. G has no proper open subgroups;
3. For each neighborhood U of e , $\cup_{n \geq 1} U^n = G$

In particular any connected Lie group is σ -compact.

Referencias

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